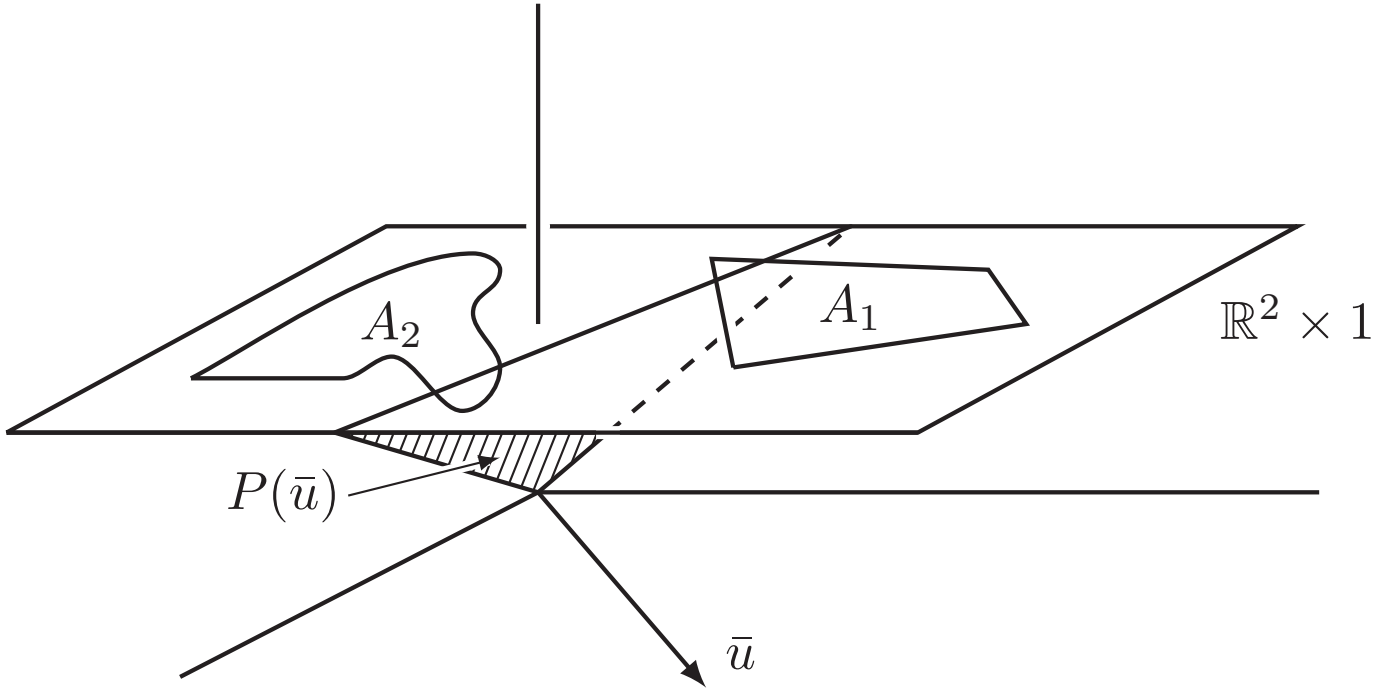

A_2
A_1
$\mathbb{R}^2 \times 1$
$P(\bar{u})$
$\bar{u}$